

Converge Bio Machine Learning Researcher - Home Assignment - External

Role: Machine Learning Researcher

AI policy: Highly Encouraged

Overview

Build a multimodal model that predicts how a mutation affects antibody–antigen binding affinity using protein sequence and 3D structure.

This is a difficult, small-data problem. We do not expect state-of-the-art performance, although strong performance is always encouraged. We care primarily about your modeling choices, training process, evaluation, error analysis, and how you would improve the model.

Antibodies and the dataset

Antibodies bind molecular targets called antigens. Their binding site is determined jointly by amino-acid sequence and 3D structure.

Use the antibody–antigen subset of [SKEMPI 2.0](#), which contains experimentally measured changes in binding free energy following mutations in structurally resolved protein complexes.

A short introduction to antibodies can be found in the [OAS paper](#). SKEMPI is described in its [original publication](#).

Models

Protein language models learn representations from amino-acid sequences, while structure models use the spatial organization and interactions between residues. A multimodal model combines information from both modalities.

You are not expected to train a foundation model or process structures at scale. Models that can be used locally or on a modest GPU include are fine, there is a wide variety of them and some include:

[ESM-2 8M/35M](#) and [ESMC 300M](#) for sequence modeling, as well as [ESMFold](#), [ESM-IF1](#), and [SaProt](#) for structure or structure-aware representations. Antibody-specific sequence and structure models such as [AbLang2](#), [IgFold](#), and [AntiFold](#) also exist.

You may use these models or any other reasonable approach.

The task

Train a multimodal sequence-and-structure model to predict mutation-induced changes in antibody–antigen binding affinity.

You may approach the task as continuous $\Delta\Delta G$ regression or as classification into high, medium, and low affinity change.

Decide how to represent the mutation, encode each modality, combine the representations, train the model, and evaluate it. Your work should include a justified method for combining the modalities, relevant model and task-performance evaluations, results on the SKEMPI benchmark, an analysis of the model's errors and limitations, and concrete proposals for what you would try next—and why.

How we evaluate

- Above all else - justifiable choices - we mostly care about why you did something, so please make sure you can stand up to your decisions.
- Modeling choices and the reasoning behind them
- Understanding of multimodal representation and fusion
- Training methodology and experimental judgment
- Quality and relevance of model evaluation
- Depth of error analysis
- Ability to identify limitations and propose strong next steps
- Code quality, clarity, and reproducibility

Submission requirements

Submit a GitHub repository containing:

- Code and environment requirements
- Instructions for running the solution
- A concise explanation of the model and key decisions
- Training and SKEMPI benchmark results
- Model evaluation and error analysis
- Limitations and proposed next steps
- AI prompt history

State the hardware and approximate runtime. Be prepared to explain and defend every part of the submission.